TABLEAU VISUALIZATIONS AND DASHBOARD DEVELOPMENT

5. Creating Visualizations in Tableau

5.1 Introduction

Tableau was used to create visual representations of the Formula 1 dataset. The main purpose of these visualizations was to make numerical information easier to understand and compare.

Charts transform raw records into graphical summaries, allowing viewers to identify differences between categories and understand the distribution of important statistics.

5.2 Bar Charts

Bar charts are useful for comparing numerical values across different categories.

In a Formula 1 analysis, bar charts can be used to compare driver points, race wins, podiums, pit-stop counts, or other selected measures.

These charts make it easier to identify higher and lower values and compare the performance of different categories.

5.3 Pie Charts

Pie charts represent the distribution of a total across different categories.

For example, the race-level weather distribution can be represented using a pie chart showing the proportions of Dry, Overcast, Light Rain, Heavy Rain, and Mixed conditions.

Tyre-compound records can also be visualized as a distribution, provided the chart clearly identifies what each value represents.

5.4 Other Performance Visualizations

The available data supports several additional types of analysis:

- Qualifying comparisons using qualifying positions and lap times.
- Lap-performance analysis using lap times, sector times, and speed measurements.
- Race-strategy analysis using pit stops, tyre information, and overtaking statistics.
- Championship comparisons using cumulative points and standings-related measures.

- Circuit analysis using circuit lengths, locations, and track characteristics.

The choice of visualization depends on the type of variable being analyzed and the objective of the comparison.

5.5 Importance of Visualization

Tableau makes it possible to present complex numerical information in a clear and structured format.

Visualizations support comparisons, reveal distributions, and make important values easier to identify. They also provide the foundation for building a dashboard that combines multiple views of the dataset.

6. Building the Interactive Dashboard

6.1 Dashboard Overview

After creating the individual worksheets and visualizations, the selected charts were brought together into a Tableau dashboard.

A dashboard combines multiple visualizations in a single interface, allowing viewers to examine different aspects of the analysis in one place.

6.2 Dashboard Design

The dashboard presents selected Formula 1 statistics in an organized layout.

Depending on the visualizations included, it can provide information about driver performance, race results, qualifying results, lap statistics, tyre compounds, weather conditions, and championship standings.

The purpose of the layout is to make comparisons easier and present the most relevant information in a single view.

6.3 Interactivity and Usability

Tableau supports interactive dashboard features such as filters, selections, and chart highlighting.

Where configured in the project, these features allow viewers to focus on selected drivers, teams, seasons, circuits, or other categories.

Clear chart titles, labels, legends, and consistent formatting help users understand the information being presented.

6.4 Dashboard Objectives

The main objectives of the dashboard are:

1. To bring important visualizations together in one place.
2. To make racing statistics easier to compare.
3. To provide a structured overview of the selected dataset.
4. To help viewers identify patterns and differences across categories.
5. To communicate the results of the analysis in an accessible format.

The Tableau dashboard serves as the central visualization component of the Formula 1 Racing Intelligence project.